

■ OBJECTS IN THE MILKY WAY

Most of the galaxy's visible material consists of stars at various stages in their life cycles: young, newly formed ones; middle-aged ones like the Sun; and older red giants and planetary nebulae. The rest is vast clouds of interstellar gas and dust.

STARS, CLUSTERS, AND NEBULAE

A star is a huge ball of hot, glowing gas that is held together by gravity. Like humans, each star is unique, differing from each other in size, brightness, color, age, and mass—that is, the amount of gas making the star.

A star's size, temperature, and color change with time. An individual star is described by the stage it has reached in its life cycle. Betelgeuse, for instance, is a red supergiant; and a planetary nebula is an old star that has pushed off its outer layer of gas.

Some stars seen from Earth are double stars. Such stars may not be related, but they look close because of our line of sight from Earth. Other doubles, such as Algol, exist together in space; stars such as this are known as binaries. Stars whose brightness varies are variable stars.

Stars form in clusters from gas and dust, and often they will drift apart. Open clusters consist of young, newly switched-on stars; globular clusters consist of older stars. A cloud of interstellar gas and dust is called a nebula. Nebulae either shine brightly or appear as a dark patch against a brighter background.



BETELGEUSE (RED SUPERGIANT)



ALGOL (BINARY)



ALDEBARAN
(RED GIANT, VARIABLE)



BUTTERFLY CLUSTER (OPEN CLUSTER)



OMEGA CENTAURI (GLOBULAR CLUSTER)



HELIX NEBULA (PLANETARY NEBULA)



LAGOON NEBULA (BRIGHT NEBULA)

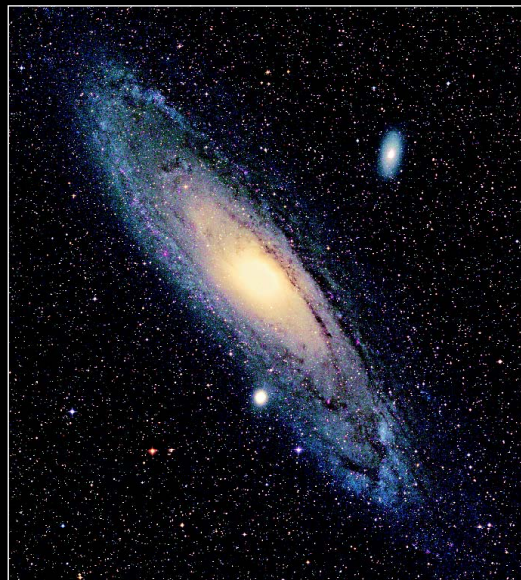


HORSEHEAD NEBULA (DARK NEBULA)

■ BEYOND THE MILKY WAY

When we look out from Earth and beyond the stars of the Milky Way, we see galaxies. They exist in every direction we look. It is estimated that there are 100–125 billion of them. Each is a vast group of stars held together by its own gravity. A single one contains billions or trillions of stars, as well as interstellar gas and dust. Galaxies vary in width from a few thousand light-years to more than a million light-years. They are so remote that even the nearest look like faint blurs of light to the naked eye.

Galaxies are classified according to shape. Spirals are disk-shaped with a central bulge and spiraling arms. A spiral galaxy with a bar-shaped center and arms winding out from those bar ends is known as a barred spiral. Ellipticals are ball-shaped, along the lines of a soccer ball, a football, a flattened ball, or something in between. Irregular galaxies have no obvious defined shape or form.

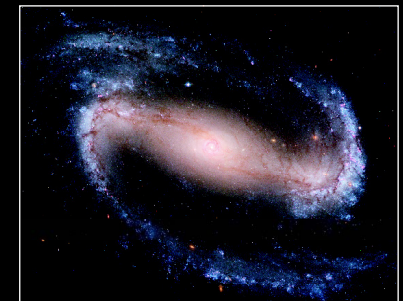


Andromeda Galaxy

The closest major galaxy to the Milky Way, Andromeda is the largest member of the Local Group. To the naked eye, this vast spiral galaxy looks like an elongated smudge of light.



LARGE MAGELLANIC CLOUD (IRREGULAR)



NGC 1300 (BARRED SPIRAL)



M87 (ELLIPTICAL)